

Hellman's Time-Memory Trade-Off (Summary Notes)

1 Objective

Given:

- Plaintext P
- Ciphertext C

Find the unknown DES key.

Hellman's idea is to reduce online search time using precomputation and additional memory.

2 Basic Idea

Instead of storing every possible key, generate long chains and store only:

(Start Point, End Point)

Intermediate values are discarded and recomputed whenever required.

3 Reduction Function

DES performs

Key $\xrightarrow{DES}$ Ciphertext

Since ciphertext may or may not be a valid key, define a reduction function

$R : \text{Ciphertext} \rightarrow \text{Key}$

4 Function

Define

$$f(K) = R(E_K(P_0))$$

where

- P_0 = fixed plaintext
- $E_K(P_0)$ = DES encryption using key K

Each iteration is

$$K_i \rightarrow E_{K_i}(P_0) \rightarrow C_i \rightarrow R \rightarrow K_{i+1}$$

Plaintext always remains P_0 .

5 Generating Chains

Choose a random starting key

$$SP = x_{i0}$$

Generate

$$x_{i1} = f(x_{i0})$$

$$x_{i2} = f(x_{i1})$$

Continue for t iterations but store only

$$SP = x_{i0} \quad EP = x_{it}$$

where

- m = number of chains
- t = chain length

6 Collisions

Example

$$1 \rightarrow 4 \rightarrow 7 \rightarrow 9$$

$$3 \rightarrow 6 \rightarrow 7 \rightarrow 9$$

Both chains merge at 7.

Effects:

- fewer unique keys covered
- wasted computation
- reduced probability of success

For this we define

$$I(x) = \begin{cases} 1, & x \text{ is new} \\ 0, & \text{otherwise} \end{cases}$$

Total unique keys

$$|A| = \sum I(x_{ij})$$

Also expected value

$$E(|A|) = \sum P(x_{ij} \text{ is new})$$

since

$$E(I) = P.$$

7 Success Probability

Coverage probability

$$P(S) = \frac{|A|}{N}$$

Using expectation,

$$P(S) = \frac{E(|A|)}{N}$$

This measures the probability that the secret key lies inside one of the generated chains.

Computing

$$P(x_{ij} \text{ is new})$$

is difficult

Instead,

$$P(x_{ij} \text{ is new}) \geq P(x_{i0}, x_{i1}, \dots, x_{ij} \text{ all new})$$

This gives a lower bound on the success probability.

By doing this over all possible ciphertexts we get equation,

$$P(S) \geq \frac{1}{N} \sum_{i=0}^{m-1} \sum_{j=0}^{t-1} \left(\frac{N - it}{N} \right)^{j+1}$$

8 Choosing Parameters

We have to choose a optimal choice to cover everything while avoiding collisions. The optimal choice

$$mt^2 = N$$

Choosing

$$m = t = N^{1/3}$$

gives

$$\text{Time} = \text{Memory} = N^{2/3}$$

instead of

$$\text{Time} = N.$$

9 Multiple Tables

One table does not cover the entire key space. Construct multiple independent tables using different reduction functions. Each additional table increases overall coverage. This idea later evolved into rainbow tables.

10 Hardware Implementation

Hellman demonstrated that the time-memory trade-off is practically feasible using dedicated DES hardware. The attack consists of an offline phase, where lookup tables are generated once using approximately N DES computations, and an online phase, where these tables are used to recover keys in about $N^{2/3}$ operations. Multiple DES processors can work in parallel, significantly reducing the attack time. Thus, the expensive precomputation is performed only once and reused for multiple future attacks.